

1 Introduction

GPS and similar systems use different reference frames for GNSS. However, the main characteristic of these reference frames is that they are all based on a geocentric Cartesian coordinate system. Consequently, when using GNSS, the coordinates of terrestrial sites are determined within their respective reference frames. Geodetic coordinates (latitude, longitude, and height) are derived from Cartesian coordinates and are more intuitive for mapping and navigation. They are based on an ellipsoidal model of the Earth, which approximates its shape more accurately than a simple sphere.

The Cartesian coordinate system used to represent locations relative to the center of the Earth, the origin (0,0,0) is located at the Earth's center of mass and the coordinates (X, Y, Z) are measured in meters from the Earth's center. The X axis is at the meridian of Greenwich at 0° longitude, while the Y axis extends from the center of the Earth to the intersection of the equator at 0° latitude. The Z axis is perpendicular to the X and Y axes, and aligns with the North and South poles of the Earth. The range of these coordinates depends on the Earth's dimensions. If we consider the radius of the Earth of 6,378 kilometers (equatorial radius), then the coordinates can range roughly from -6,378,000 meters to +6,378,000 meters along each axis.

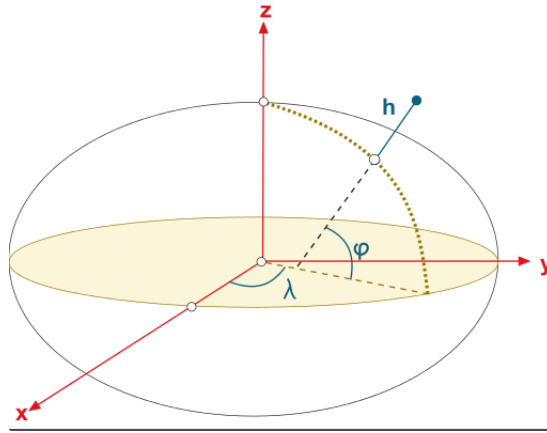


Figure 1: Ellipsoidal and Cartesian Coordinates conversion.
Source: European Space Agency (ESA) [2].

The (X, Y, Z) Cartesian coordinates can be expressed in the ellipsoidal coordinates (ϕ, λ, h) , where ϕ and λ are, respectively, the latitude and longitude from the ellipsoid, and h is the height above the ellipsoid.

The ellipsoidal coordinate system is a three-dimensional orthogonal coordinate system used to represent points on the Earth's surface. The system is based on an ellipsoid, which is a mathematically defined surface that approximates the shape of the Earth more accurately than a sphere. The latitude (ϕ), is the angle between the equatorial plane and a line that passes through a point on the Earth's surface and the center of the Earth, it takes values from -90° at the South Pole to $+90^\circ$ at the North Pole. While the longitude (λ) is defined as the angle between the prime meridian and a line that passes through a point on the Earth's surface and the center of the Earth and goes from -180° to $+180^\circ$, with 0° at the prime meridian. The third component, h represents the height above the ellipsoid. This can be also represented as the sum of the orthometric height or height above sea level (H) and the geoid height (N). The orthometric height (H) is the height above an imaginary surface called the geoid, which is determined by the earth's gravity and it is an approximation of the

geoid. By definition, the geoid is an equipotential surface in the Earth's gravity field, it describes the irregular shape of the earth and is the true zero surface for measuring elevations [1]. The height can vary depending on the point's elevation above or below the ellipsoid. It can be positive (above the ellipsoid) or negative (below the ellipsoid).

The transformations from the ellipsoidal coordinates into Cartesian are equally possible.

And developing conversion routines from geodetic to Cartesian coordinates and vice versa are the goals of the first part of this assignment. Given the data of latitude, longitude, orthometric and geoid height, and parameters of the ellipsoid, the transformations will be performed. At the same time that some verifications will be done from the given Cartesian coordinates of a position given.

But surveyors typically do not need the coordinates of terrestrial points in a global frame; instead, they prefer results in a local coordinate frame, such as geodetic or ellipsoidal coordinates, plane coordinates, or vectors combined with other terrestrial data. Since GNSS systems use global reference frames, transforming these coordinates into local frames is often necessary for practical applications. [4]

Global Cartesian coordinates (like those in the WGS-84 reference frame) are necessary for applications that require a consistent global reference, such as satellite navigation and global mapping. Local Cartesian coordinates, on the other hand, are used for more localized tasks, such as surveying and construction, where the focus is on a specific area.

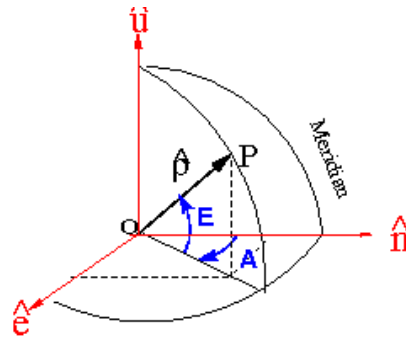


Figure 2: ENU and Cartesian Coordinates conversion.

Source: European Space Agency (ESA) [3].

The ENU (East-North-Up) coordinate system provide a straightforward way to represent positions and movements relative to a specific location. The system is based on a plane tangent to the Earth's surface at a specific location. This local tangent plane is practical for applications that require precise local positioning, such as surveying and tracking. The East (E) coordinate is given in the axis that points towards the east, tangent to the Earth's surface. The North (N) coordinate is in the axis that points towards the north, also tangent to the Earth's surface, and Up (U) is in the axis that points upwards, perpendicular to the Earth's surface.

GNSS systems often provide data in global Cartesian coordinates. Transforming these coordinates into ENU makes it easier to integrate GNSS data with local measurements and perform accurate geospatial analysis.

In the second part of this exercise, the aim is to determine the satellites visible from a point on the Earth's surface by converting geocentric or global Cartesian coordinates into a local Cartesian

coordinate system (e,n,u) and calculating the azimuths, distances and elevations of the satellites to achieve our aim.

Azimuth and elevation are measures used to determine the position of a satellite passing overhead. While the azimuth tells you in which direction to look, the elevation angle tells you how high in the sky to look. In addition, if this angle is between $-\pi/2$ and $\pi/2$, the satellite will be visible from a point on the Earth's surface.

2 Methodology

In order to achieve the goals in the assignment previously presented, the expressions and the routines necessary for the transformations between coordinates systems should be introduced.

From Ellipsoidal to Cartesian coordinates

The Cartesian coordinates of a point (X, Y, Z) can be obtained from the ellipsoidal coordinates (ϕ, λ, h),

$$\begin{aligned} X &= (N + h) \cos \phi \cos \lambda, \\ Y &= (N + h) \cos \phi \sin \lambda, \\ Z &= (N + h - e^2 \cdot N) \sin \phi. \end{aligned}$$

Where e^2 represents the first numerical eccentricity,

$$e^2 = \frac{a^2 - b^2}{a^2},$$

and the radius of curvature in the prime vertical is

$$N = \frac{a^2}{\sqrt{a^2 \cos^2(\phi) + b^2 \sin^2(\phi)}}.$$

In this expressions, a is the semi-major axis of the ellipsoid and b is the semi-minor axis, which is easily computed from the parameters of the ellipsoid GRS80, flattening (f) and the semi-major axis,

$$f = \frac{(a - b)}{a}.$$

This expressions are introduced in a Matlab function `geodetictocartesian`, that from the ellipsoidal coordinates (ϕ, λ, h), and the ellipsoid GRS80 parameter, flattening (f) and the semi-major axis (a), the Cartesian coordinates (X, Y, Z) will be returned. And as previously explained, the height above the ellipsoid h , will be obtained from the sum of the orthometric height or height above sea level (H) and the geoid height (N) given. The created function will transform the latitude and longitude values from degrees to radians, with the help of the built in function in Matlab `deg2rad`, and will compute all the calculations in radians.

To check the coordinates obtained through the transformations, the error between the reference coordinate values and the transformed coordinates, is computed as the difference between them.

From Cartesian to Ellipsoidal coordinates

The ellipsoidal coordinates of a point (ϕ, λ, h), can be obtained from the cartesian coordinates (X, Y, Z) following the process explained in [4]:

1. Compute the longitude as $\tan \lambda = \frac{Y}{X}$.
2. Compute the radius of a parallel, p , as $p = \sqrt{X^2 + Y^2}$.
3. Compute an approximate value of latitude $\phi_{(0)}$ from $\tan \phi_{(0)} = \frac{Z}{p \cdot (1 - e^2)}$. Where e^2 is the first numerical eccentricity.

4. Compute the approximate value of radius of curvature in the prime vertical, $N_{(0)}$, as

$$N_{(0)} = \frac{a^2}{\sqrt{a^2 \cos^2(\phi_{(0)}) + b^2 \sin^2(\phi_{(0)})}}.$$

5. Compute the ellipsoidal height by $h = \frac{p}{\cos(\phi_{(0)})} - N_{(0)}$.
6. Compute an improved value for the latitude by $\tan \phi = \frac{Z}{p \cdot (1 - e^2 \frac{N_{(0)}}{N_{(0)} + h})}$.
7. The process is repeated while ϕ and $\phi_{(0)}$ are different.

As before, a Matlab function `cartesiantogeodetic` is created to implement this process. From the Cartesian coordinates (X, Y, Z), and the ellipsoid GRS80 parameter, flattening (f) and the semi-major axis (a), the ellipsoidal coordinates (ϕ, λ, h) will be computed. The created function will return the values of this coordinates in degrees.

From Cartesian geocentric to ENU coordinates

The ENU coordinates (e, n, u) are obtained by applying the following transformation:

$$\begin{bmatrix} e \\ n \\ u \end{bmatrix} = R_e^n \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix}$$

Where R is the rotation matrix

$$R_e^n = R_x(90^\circ - \phi_a) \cdot R_z(90^\circ + \lambda_a) = \begin{bmatrix} -\sin \lambda_a & \cos \lambda_a & 0 \\ -\sin \phi_a \cos \lambda_a & -\sin \phi_a \sin \lambda_a & \cos \phi_a \\ \cos \phi_a \cos \lambda_a & \cos \phi_a \sin \lambda_a & \sin \phi_a \end{bmatrix}.$$

The rotation matrix is obtained as the product of the rotation matrix in the x axis and in the z axis. This matrices are:

$$R_x(\theta_1) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_1 & \sin \theta_1 \\ 0 & -\sin \theta_1 & \cos \theta_1 \end{bmatrix}; R_z(\theta_2) = \begin{bmatrix} \cos \theta_2 & \sin \theta_2 & 0 \\ -\sin \theta_2 & \cos \theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Where $\theta_1 = 90^\circ - \phi_a$ and $\theta_2 = 90^\circ + \lambda_a$.

The last term in the transformation is given by

$$\begin{aligned} \Delta X &= X_{SV} - X_a \\ \Delta Y &= Y_{SV} - Y_a \\ \Delta Z &= Z_{SV} - Z_a \end{aligned}$$

which represents the differences between the cartesian coordinates of the observer, given by the subindex a, and the satellites, indicated by the subindex SV.

We will compute for each satellite the ENU coordinates.

Once the transformation of the coordinates is done, we can compute the azimuth, the elevation angle, the zenith angle and the distance to each satellite.

$$\text{Azimuth, } \alpha = \tan^{-1} \left(\frac{e}{n} \right),$$

$$\text{Elevation angle, } \textit{elevation} = 90^\circ - z,$$

$$\text{Zenith angle, } z = \cos^{-1} \left(\frac{u}{s} \right),$$

$$\text{Distance, } s = \sqrt{e^2 + n^2 + u^2}.$$

A Matlab function `enucoordinates` was created to obtain the ENU coordinates, the azimuths, the distances, the zenith angles and the elevation angles from the cartesian coordinates of the observer and the cartesian coordinates of the satellites. Finally, with the azimuth values, we will determine the satellites that are visible by checking which azimuths values are positive.

3 Results

To present the results obtained, we will divide this section in two parts, one for each task of the assignment.

Part A. Geodetic and Cartesian coordinates.

The position of a point in geodetic coordinates were given, the latitude is $90^{\circ}20'59''$, longitude is $18^{\circ}04'10''$ and the ellipsoidal height is 85 meters, obtained from the sum of 62 meters of orthometric height and 23 meters of geoid height. Using the parameter semi-major axis and flattening of the ellipsoid GRS80, when we transform the coordinates into Cartesian coordinates, we obtain

$$(X, Y, Z) = (3098917.24172898, 1011053.41132697, 5463972.35267995).$$

If we compare this coordinates with the given reference cartesian coordinates, we obtain that the difference between our computations and the reference coordinates are

$$\begin{aligned}\Delta X &= X - X_{ref} = 0.00172897987067699 \text{ meters,} \\ \Delta Y &= Y - Y_{ref} = 0.00132697040680796 \text{ meters,} \\ \Delta Z &= Z - Z_{ref} = 0.00267994683235884 \text{ meters.}\end{aligned}$$

The values of the transformed coordinates and the reference coordinates are less than 3 centimeters, a difference small enough to consider the transformed coordinates to be sufficiently accurate.

To check the transformation process between Cartesian coordinates and ellipsoidal coordinates, we calculate the difference between the coordinates obtained by applying the transformation and the original coordinates given, and we obtain values very close to zero, which means that the process is very accurate.

We should also check if the process works correctly for the dangerous positions. In this positions that are interested to be study, we should consider positions in the Equator, where the latitude is 0° , in this positions the transformation works because at the equator, the sine term for latitude becomes zero ($\sin(0) = 0$), and this means that we obtain simpler formulas, also we do not have a moment where we need to divide by $\sin(0)$.

The Poles, were the latitude take values of $\pm 90^{\circ}$, we have that $\cos(\pm 90^{\circ}) = 0$, as before this could give us more simpler expressions and when computing the radius of curvature in the prime vertical, it is well defined.

We should consider points where they have latitude or longitude $\pm 180^{\circ}$, it behaves correctly by the same reason as for Equator and the poles.

In the transformations between the, Cartesians and the elliptic coordinates, we should consider the cases where we compute longitude from $\tan \lambda = Y/X$. In this cases we risk to have a division by 0, but when doing the computation of the longitude the build in function in Matlab `atan2d`, made the code robust to this cases.

Part B. Global and Local Cartesian coordinates.

For the 29 satellites whose Cartesian coordinates where given, we have compute the ENU coordinates, the azimuth, the elevation angle and the distance from the point that represents the origin of

the system. This point given in geodetic coordinates is $(90^{\circ}20'59'', 18^{\circ}04'10'', 85)$. Since this coordinates are elliptic, we transform them using the method explained for the previous section, and we obtain the Cartesian coordinates. We should also consider that the data given for the satellites, are the Cartesian coordinates given in kilometers, since for the computations, we need this coordinates in meters, we should multiply the values in kilometers by 1000.

To check that the proposed routine works correctly, we have introduced a point whose coordinates are $(90^{\circ}20'59'', 18^{\circ}04'10'', 185)$, i.e. it has the same latitude and longitude, but an ellipsoidal height of 100 meters above our reference point. If we calculate the ENU coordinates for this point, we get $(0, 0, 100)$. This means that if we take a point above the point that is our center in the reference system, our coordinates north and east are zero, that is, they are at the origin, but the coordinate up is 100, so this point is 100 meters above the origin on the z-axis (u), but at the origin on the x-axis (e) and y-axis (n).

Once the routine has been checked and the coordinates, azimuth, elevation and distances have been calculated, we should check whose satellites are visible from the point on the Earth's surface that is the origin of our local system. This can be done by checking the elevation angle; if this angle is possible, the satellites are above the horizon, i.e. they are visible.

Checking this angle, we find that only 11 of the 29 satellites are visible from the selected point. In the table 1 we have included the calculated information for the 11 visible satellites.

Satellite Number	East	ENU (m)		Elevation ($^{\circ}$)	Azimuth ($^{\circ}$)	Slant Distance (m)
		North	Up			
1	8.9852e+06	-1.7319e+07	1.1745e+07	31.046	152.58	2.2773e+07
3	2.9705e+06	-2.3680e+07	5.6301e+06	13.274	172.85	2.4521e+07
7	-1.4022e+07	2.1901e+07	4.9696e+05	1.0948	327.37	2.6010e+07
9	9.9475e+06	2.0926e+07	5.7070e+06	13.837	25.425	2.3862e+07
11	-1.1266e+07	-1.7231e+05	1.7639e+07	57.431	269.12	2.0931e+07
14	1.3550e+07	-5.3374e+06	1.5905e+07	47.522	111.50	2.1566e+07
19	-2.1802e+06	-1.5869e+07	1.4869e+07	42.871	187.82	2.1856e+07
20	-1.9720e+07	-1.3881e+07	4.7106e+06	11.053	234.86	2.4572e+07
22	1.7714e+07	7.6451e+06	1.2069e+07	32.029	66.656	2.2757e+07
28	-1.9015e+07	1.0569e+07	9.0152e+06	22.509	299.07	2.3549e+07
31	-1.5559e+07	1.5427e+07	8.1048e+06	20.300	314.76	2.3362e+07

Table 1: Satellite positional data in ENU coordinates, elevation angles, azimuth angles, and slant distances.

The image 3 include a representation in 2D of the visible satellite and the image 4 is a representation in 3D of the satellites and the Earth.

The 2D representation or skyplot, take the values of the azimuths and the elevation angles and take as origin of coordinates a position in the origin of the local coordinate system. As for the 3D, we take the Cartesian coordinates of the satellites and we represent the Earth in whose surface we include the point that will be the origin in the local system, making easy to see which of the satellites will be observables from that point.

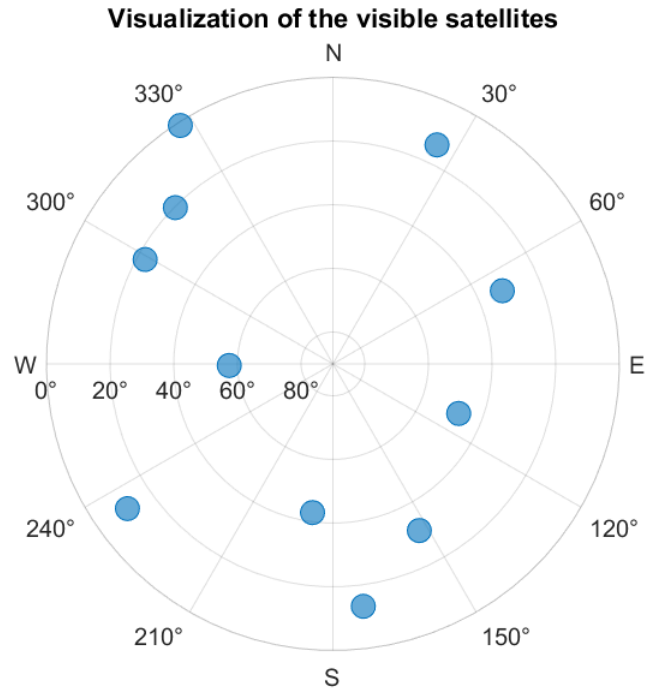


Figure 3: Skyplot of the visible satellites.

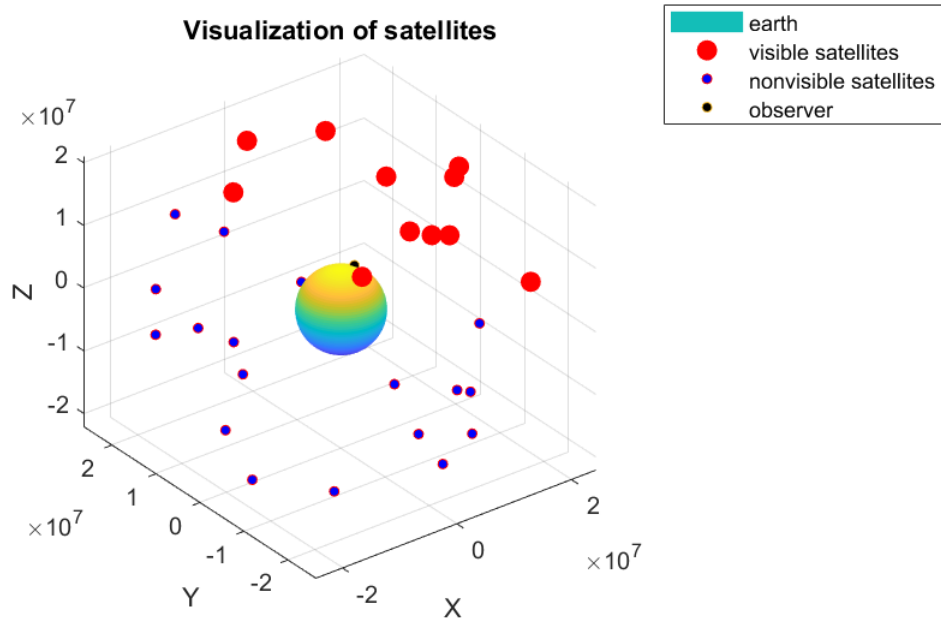


Figure 4: 3D representation of the satellites and the Earth.

4 Analysis and Discussion

The results obtained seem to be accurate for the transformation of coordinates between Cartesian and geodetic coordinates, since the differences obtained between the reference coordinates and the transformed coordinates are small enough not to misidentify the position. It is important to mention that the cartesian to geodetic transformation, has given us less deviation than the transformation in the other direction, this could be due to the unit of measurement of the coordinates.

At the same time, the transformation methods are robust even for difficult positions, where the trigonometric functions can work even if there are divisions by zero. Even for other difficult positions, such as the equator or the poles, the routines still return adequate values. It is important to mention that it is no problem to work with positive or negative coordinates.

Because of the small deviation between the reference coordinates and the transformation coordinates, and the ability to work well at difficult points, we conclude that the code is accurate and robust.

As for the local Cartesian coordinate system, the test done with a point located 100 meters above the origin of coordinates, has prove that the method works correctly, since for the latitude and the longitude the point was located in the origin, but it was deviated in the z-axis or up-axis 100 meters, which was the difference between the reference point on the Earth's surface and the point located above it.

For the satellites, the values of distances, azimuths and elevation angles seems to be correct and as for the number of visible satellites, if we check the 3D representation, if we draw a plane that intersect the Earth's surface in the point that is the origin of the local system, only the 11 satellites with positive elevation angle will be the ones above that plane.

References

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